

COURSE SYLLABUS

Advanced SQL Server Performance Tuning

COURSE NUMBER

65318A

DURATION

2 days · 6 modules

AUDIENCE LEVEL

Advanced

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This 2-day course builds on the SQL Server 2025 query analysis skills taught in 65317A and focuses on real-world troubleshooting techniques. Students will be taught to tune and optimize queries using SSMS 22 and other tools on SQL Server 2025.

AT COURSE COMPLETION

After completing this course, you will be able to:

Analyze query execution plans to identify performance bottlenecks in SQL Server 2025

Apply real-world tuning techniques using SSMS 22 and advanced diagnostic tools

Optimize query performance through index strategy and execution plan optimization

Troubleshoot and resolve performance issues in production SQL Server environments

Leverage SQL Server 2025 new features and Microsoft tools for continuous performance monitoring

PREREQUISITES

- Course 65317A - Querying and Analyzing SQL Server 2025, or equivalent knowledge
 - Solid understanding of T-SQL and query fundamentals
 - Experience working with SQL Server in production environments
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COURSE OUTLINE

MODULE 1 Performance Troubleshooting Methodology

140 min

Establish a systematic approach to identifying and diagnosing SQL Server performance problems. Learn to establish performance baselines and apply a consistent troubleshooting workflow.

LESSONS

- Lesson 1:** Introduction to SQL Server Performance Troubleshooting
- Lesson 2:** Identifying Performance Problems
- Lesson 3:** Establishing Performance Baselines
- Lesson 4:** Systematic Troubleshooting Workflow
- Lesson 5:** Workload Analysis Techniques
- Lesson 6:** SSMS 22 Diagnostic Features

LAB EXERCISES

- Lab 1 – Identifying a Slow Query

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Identify common SQL Server performance problems using diagnostic queries and tools
- Establish performance baselines for proactive monitoring
- Apply a systematic troubleshooting workflow to diagnose performance issues
- Analyze workload statistics to locate inefficient queries

MODULE 2 Advanced Index Tuning

140 min

Master index optimization strategies including missing index identification, maintenance strategies, and fragmentation management to improve query performance.

LESSONS

- Lesson 1:** Index Architecture and Performance Impact
- Lesson 2:** Missing Index Analysis with DMVs
- Lesson 3:** Advanced Index Design Strategies
- Lesson 4:** Index Fragmentation Analysis
- Lesson 5:** Index Maintenance Strategies
- Lesson 6:** Testing and Validating Index Improvements

LAB EXERCISES

- Optimizing Query Performance Through Index Analysis and Tuning

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Identify missing indexes using SQL Server 2025 recommendations and DMVs
- Design and implement optimized index strategies for complex workloads
- Evaluate index fragmentation and apply appropriate maintenance strategies
- Test and validate query performance improvements after index optimization

MODULE 3 Query Store Performance Diagnostics

140 min

Leverage Query Store capabilities in SQL Server 2025 to analyze historical query performance, identify regressions, and stabilize execution plans.

LESSONS

Lesson 1: Query Store Architecture and Components

Lesson 2: Configuring Query Store for Production Workloads

Lesson 3: Query Store Reports in SSMS
22

Lesson 4: Identifying and Analyzing Query Performance Regressions

Lesson 5: Plan Forcing Strategies and Implementation

Lesson 6: Query Store Maintenance and Best Practices

LAB EXERCISES

Query Store Performance Diagnostics and Plan Forcing

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Describe Query Store architecture and configuration options in SQL Server 2025

Analyze historical query performance data to identify trends and regressions

Identify queries with performance regressions using Query Store reports

Force optimized execution plans to stabilize query performance

MODULE 4 Wait Statistics and Resource Bottlenecks

140 min

Diagnose system-level performance problems by interpreting wait statistics and identifying CPU, I/O, and memory bottlenecks in SQL Server 2025.

LESSONS

Lesson 1: Understanding Wait Statistics

Lesson 2: Querying and Interpreting Wait Statistics

Lesson 3: Diagnosing CPU Bottlenecks and Parallelism

Lesson 4: I/O Bottlenecks and Disk Subsystem Analysis

Lesson 5: Memory Pressure and Buffer Pool Issues

Lesson 6: DMVs for Resource Monitoring

LAB EXERCISES

Diagnosing Resource Bottlenecks Using Wait Statistics

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Interpret wait statistics to identify the root cause of performance problems

Diagnose CPU bottlenecks and implement mitigation strategies

Identify and resolve I/O bottlenecks affecting query performance

Recognize and address memory pressure issues in SQL Server 2025

Troubleshoot concurrency issues including blocking chains, deadlocks, and isolation level impacts on SQL Server 2025 workloads.

LESSONS

Lesson 1: Understanding SQL Server Locking Mechanisms

Lesson 2: Blocking Chains and Detection Strategies

Lesson 3: Deadlocks: Detection, Analysis, and Resolution

Lesson 4: Isolation Levels and Concurrency Impact

Lesson 5: Advanced Concurrency Optimization Techniques

Lesson 6: Monitoring and Troubleshooting Concurrency Issues

LAB EXERCISES

Diagnosing and Resolving Blocking and Deadlocks

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Identify and analyze different lock types and their impact on concurrency

Diagnose and resolve blocking chains affecting transaction throughput

Capture and analyze deadlock graphs using SQL Server 2025 tools

Apply appropriate isolation levels to balance concurrency and consistency

MODULE 6 Advanced Query Optimization

140 min

Master advanced optimization techniques including parameter sniffing troubleshooting, plan stability, memory grants, and adaptive query processing in SQL Server 2025.

LESSONS

- Lesson 1:** Understanding Parameter Sniffing and Plan Reuse
- Lesson 2:** Detecting and Diagnosing Parameter Sniffing Issues
- Lesson 3:** Parameter Sniffing Mitigation Strategies
- Lesson 4:** Memory Grants and Performance Impact
- Lesson 5:** Adaptive Query Processing in SQL Server 2025
- Lesson 6:** Query Hints and Plan Guides for Plan Stability

LAB EXERCISES

- Diagnosing and Resolving Parameter Sniffing and Memory Grant Issues

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Diagnose and resolve parameter sniffing issues affecting query performance
- Implement strategies for maintaining plan stability in dynamic workloads
- Analyze and optimize memory grant issues causing performance problems
- Leverage adaptive query processing features in SQL Server 2025

BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3

SKILLS TO BE INTEGRATED

Analyze workload statistics to locate inefficient queries and diagnose bottlenecks

Evaluate index recommendations, design optimized indexes, and test query performance improvements

Analyze historical query performance using Query Store, identify performance regressions, and force optimized execution plans

DAY 2 CAPSTONE Modules 4, 5, 6

SKILLS TO BE INTEGRATED

Analyze wait statistics, identify resource constraints (CPU, I/O, memory), and diagnose system bottlenecks

Simulate blocking transactions, capture deadlock graphs, and resolve concurrency issues

Observe plan reuse problems, test parameter sniffing scenarios, and implement mitigation techniques

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.